Jayraj Kiran Ladkat

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Education

10/2022 – 10/2026 **B.Tech. in Computer Science and Engineering** MIT World Peace University
Pune, Maharashtra, India

Research Interests

AI for Education, LLM/VLM Evaluation, Evolutionary Prompt Optimisation, and ML Systems

Research Experience

Research Intern 07/2026 – Present
TIGER-Lab, University of Waterloo *Waterloo, Canada (Remote)*

- Advisor: Max Ku.
- Developing an automated multi-step generation pipeline for explanatory educational videos (script → code → render → narration).
- Optimising system prompts using evolutionary prompt optimisation methods to dynamically encode user preferences and align pedagogical style (step-by-step intuition vs. mathematical depth). *Work in progress.*

Publications

P2. Shreya Mandekar, **Jayraj Kiran Ladkat**, Madhura Phatak, and Bhavana Tiple, "Co-Adaptation over Allocation: KL-Guided Mixed-Precision Quantization-Aware Training," *accepted poster*, IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC), 2026. [Acceptance Proof]

P1. **Jayraj Kiran Ladkat** and Pranati Waghodekar, "Real-World Scalability of PostgreSQL: Practical Techniques Use Case Study," in *IEEE International Conference on Blockchain and Distributed Systems Security* (ICBDS 2025). [IEEE Xplore 11377486]

Professional Experience

Engineering Intern 08/2025 – 12/2025
Publicis Sapient *Gurugram, Haryana*

- Optimised an AI agent data-generation pipeline: reduced per-run token usage, redesigned the help interface, and improved synthetic dataset quality.
- Extended an AI recruitment screening tool with request prefetching, responsive multi-device frontend, and structured assessment logic—cutting evaluation turnaround and manual HR review effort.

Software Engineering Intern 06/2025 – 08/2025
Elixir AI *Pune, Maharashtra*

- Built features for the *Climate Smart Public Health Platform*, a Harvard University–Malagasy Ministry of Public Health collaboration.
- Developed the full-stack geospatial application end-to-end: Flask + PostgreSQL/PostGIS backend, React map/dashboard frontend, Docker-based local dev. Demonstrated the platform to Harvard and Ministry stakeholders.

Cloud & Full-Stack Intern 01/2025 – 02/2025
Centre for Development of Advanced Computing (C-DAC) *Pune, Maharashtra*

- Introduced NATS-based asynchronous messaging into the service architecture, reducing request wait time under load.

Selected Projects

Dexterio — Autonomous Educational Animation Generation Engine 2026

- Engineered an Agentic RAG pipeline orchestrating multi-step lesson storyboard planning and modular Manim animation code generation grounded in course materials.
- Architected a distributed worker fleet using Apache Kafka, Daytona/Docker remote VM sandboxes for safe code compilation, and PostgreSQL-backed LangGraph checkpointers for stateful thread resumption.

StudEx — RAG Tutors for Core CS Subjects 2024 – 2025

- Built an educational platform with subject-specific chatbots (Theory of Computation, ML, Blockchain, Cloud) grounded in course PDFs via OpenAI embeddings and Pinecone vector store (LangChain orchestration).
- Designed a responsive Next.js/TypeScript dashboard with animated expandable cards (Framer Motion), Prisma ORM persistence, and transactional email via React Email.

Website Builder — In-Browser Full-Stack AI Development Platform 2024

- Developed an AI-powered browser-based platform converting natural language prompts into functional multi-file web applications using React, TypeScript, and the WebContainer API.
- Implemented real-time dynamic code execution, a syntax-highlighted editor, virtual file explorer, and client-side instant preview rendering without remote server dependencies.

Demex — Full-Stack Search Engine 2024

- Built a robots.txt-compliant web crawler in TypeScript that indexes content and stores web graph topologies in PostgreSQL.
- Implemented ranking combining TF-IDF lexical scoring and PageRank; built a Next.js search UI with real-time query completion and rich result previews.

One Stop Deployment — Automated Microservice CI/CD Platform 2024

- Developed a distributed one-click deployment engine for GitHub repositories using containerised Docker build workers and AWS ECS/S3.
- Implemented real-time log streaming using Apache Kafka and a reverse-proxy system for dynamic subdomain routing, providing instant preview URLs.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, C/C++

ML / AI: LLM/VLM evaluation; Agentic RAG (OpenAI embeddings, Pinecone, LangChain); evolutionary prompt optimisation; mixed-precision QAT; PyTorch

Systems: PostgreSQL/PostGIS, Apache Kafka, NATS, Docker, AWS (ECS, S3), WebContainers, Flask, Node.js, React, Next.js

Tools: Git, REST APIs, Prisma ORM, Drizzle ORM, LangGraph